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2.4.

13) $x(n) = \delta(n-10)$

$$\begin{aligned} \text{DFT}[x(n)] &= \sum_{n=0}^{N-1} \delta(n-10) e^{-j\frac{2\pi}{N}kn} \\ &= e^{-j\frac{2\pi}{N}k \cdot 10} \quad k=0, \dots, N-1 \end{aligned}$$

15) $x(n) = e^{j\frac{2\pi}{N}mn}$

$$\begin{aligned} \text{DFT}[x(n)] &= \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}mn} e^{-j\frac{2\pi}{N}kn} \\ &= \sum_{n=0}^{N-1} e^{j\frac{2\pi}{N}(m-k)n} \\ &= \frac{1 - e^{j\frac{2\pi}{N}(m-k)N}}{1 - e^{j\frac{2\pi}{N}(m-k)}} \end{aligned}$$

$$= \begin{cases} N & k=m \\ 0 & k \neq m \end{cases} \quad k=0, \dots, N-1$$

17)

$$x(n) = e^{j\omega_0 n} p_N(n)$$

$$\text{DFT}[p_N(n)] = \sum_{n=0}^{N-1} e^{-j\frac{2\pi}{N}kn}$$

$$\begin{aligned} \text{DFT}[x(n)] &= \sum_{n=0}^{N-1} e^{j\omega_0 n} p_N(n) W_N^{kn} \\ &= \sum_{n=0}^{N-1} e^{j\omega_0 n} e^{-j\frac{2\pi}{N}kn} \end{aligned}$$

$$= \sum_{n=0}^{N-1} e^{j(\omega_0 - \frac{2\pi}{N}k)n} = \frac{1 - e^{j(\omega_0 - \frac{2\pi}{N}k)N}}{1 - e^{j(\omega_0 - \frac{2\pi}{N}k)}} \quad k=0, \dots, N-1$$



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90)

$$x(n) = n P_N(n)$$

$$X(k) = \sum_{n=0}^{N-1} n P_N(n) W_N^{kn}$$

$$\text{由 } x(n) = x((n-1)) P_N(n) + N \delta(n) = P_N(n)$$

其DFT为

$$X(k) = X(k) W_N^k + N = N \delta(k)$$

$$\Rightarrow X(k) = \frac{N \delta(k) - N}{1 - W_N^k} = \frac{-N}{1 - W_N^k} \quad k=1, \dots, N-1$$

$$k=0 \text{ 时 } X(0) = \sum_{n=0}^{N-1} n = \frac{N^2 - N}{2}$$

$$X(k) = \begin{cases} \frac{N^2 - N}{2} & k=0 \\ \frac{-N}{1 - W_N^k} & k=1, \dots, N-1 \end{cases}$$



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3.7.

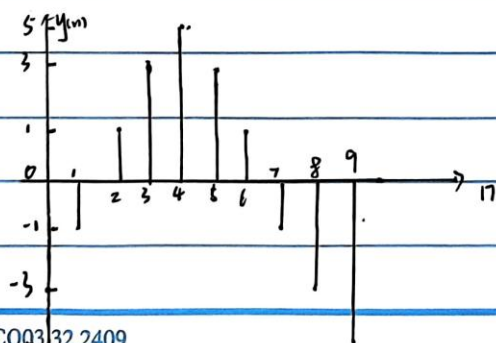
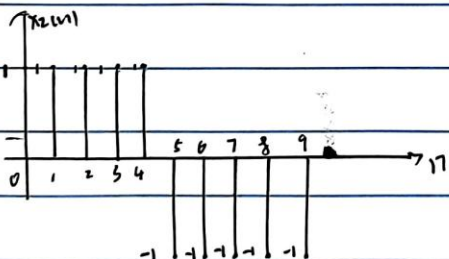
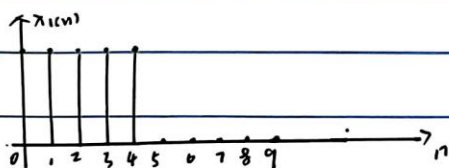
$$x_1(n) = \{1, 1, 1, 1, 1, 0, 0, 0, 0, 0\}$$

$$x_2(n) = \{1, 1, 1, 1, 1, -1, -1, -1, -1, -1\}$$

$$y_1(n) = x_1(n) \otimes x_2(n) = \{1, 2, 3, 4, 5, 3, 1, -1, -3, -5, -4, -3, -2, -1, 0, 0, 0, 0\}$$

$$y(n) = y_1(n) \downarrow 2$$

$$= \{-3, -1, 1, 3, 5, 3, 1, -1, -3, -5\}$$





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3-4

$$X(k) = \text{DFT}[x(n)] = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$\text{DFT}[x(n)] = \sum_{n=0}^{N-1} x(n) W_N^{kn}$$

$$\text{其中 } X(n) = \sum_{m=0}^{N-1} x(m) W_N^{km}$$

$$\begin{aligned} \text{则 } \text{DFT}[X(k)] &= \sum_{k=0}^{N-1} \sum_{m=0}^{N-1} x(m) W_N^{nm} W_N^{kn} \\ &= \sum_{m=0}^{N-1} x(m) \sum_{n=0}^{N-1} W_N^{n(m+k)} \end{aligned}$$

$$\sum_{n=0}^{N-1} W_N^{n(m+k)} = \begin{cases} N, & m = N-k \\ 0, & m \neq N-k \end{cases}$$

$$= N \delta(m - N + k)$$

$$N \sum_{m=0}^{N-1} x(m) \delta(m - N + k)$$

$$= N x(N-k)$$

$$\therefore \text{DFT}[X(k)] = N x(N-k) \quad k=0, \dots, N-1$$



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3-5.

$$x(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn}$$

由 DFT 唯一性

$$x(n) = \frac{1}{N} \sum_{k=0}^{N-1} x(k) W_N^{-kn}$$

$$X(w) = \sum_{k=0}^{N-1} x(k)$$

3-7.

(1) $\because x(n)$ 为实序列.

证明

$$x(k) = \sum_{n=0}^{N-1} x(n) e^{-j\frac{2\pi}{N}kn}$$

$$= \sum_{n=0}^{N-1} x(n) [\cos(\frac{2\pi}{N}kn) - j \sin(\frac{2\pi}{N}kn)]$$

$$= \sum_{n=0}^{N-1} x(n) \cos(\frac{2\pi}{N}kn) - j \sum_{n=0}^{N-1} x(n) \sin(\frac{2\pi}{N}kn)$$

$\text{Re}[x(k)]$ 为偶函数 $\text{Im}[x(k)]$ 为奇函数

证明 $x(k) = x^*(N-k)$ 具有共轭对称性

(2) $\because x(n)$ 为实序列, 且 $x(n) = x(N-n)$

$$\therefore x(n) = x^*(N-n)$$



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$x(n)$ 为实共轭对称序列.

$$\text{Re}[x(n)] = x(n)$$

$$\text{DFT}[x(n)] = \text{DFT}[x(n)] = \text{Re}[X(k)] = X(k)$$

$\therefore X(k)$ 为实序列, 且共轭对称

$$\therefore X(k) = X(N-k)$$

(3) $\therefore x(n)$ 为实序列

$$\therefore X(k) = X^*(N-k)$$

$$\text{又} \because x(n) = -x(N-n)$$

$$\therefore X(k) = -X^*(N-k), X(k) = X_0(k)$$

$$\text{DFT}[x(n)] = \text{DFT}[x_0(n)] = j \sin[k\omega_0] = j X(k)$$

$\therefore$ ~~$X(k)$~~ $\text{DFT}[x(n)]$ 为纯虚函数, 且奇对称



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3.9.

$$\begin{aligned} Y(k) &= \sum_{n=0}^{MN-1} x(n) w_{MN}^{kn} \\ &= \sum_{n=0}^{N-1} x(n) w_{MN}^{kn} \end{aligned}$$

$$= x(n) \quad \text{其中 } k=0, \dots, MN-1.$$

3.14.

$x(n)$ 在 $n=0, \dots, 7$ 处有值

$y(n)$ 在 $n=0, \dots, 19$ 处有值

用时域卷积和循环卷积关系可得: 令 $y'(n) = x(n) \times y(n)$

$$f(n) = \text{DFT}[x(n)] = y'(n)_{20 \text{ 点 DFT}}$$

$y'(n)$ 长度为 27. $n=0, \dots, 26$

$$\therefore L=20 < 27.$$

$$\therefore f(n) \neq y'(n)$$

$f(n)$ 与 $y'(n)$ 在 $7 \leq n \leq 19$ 处相等



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3-15.

m : $x(n)$ 为实序列

$$\therefore x(k) = x^*(N-k)$$

$$\therefore x(5) = x^*(3) = 0.125 + j0.658$$

$$x(6) = x^*(2) = 0$$

$$x(7) = x^*(1) = 0.125 + j0.3018$$

$$12) \quad x_1(n) = x((n+5))_8 R_8(n)$$

$$x_1(k) = W_8^{-5k} x(k)$$

13)

$$\therefore x_2(n) = x(n) e^{j\frac{\pi}{4}n}$$

$$\therefore x_2(k) = x((k-1))_8 R_8(n)$$



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3-18

已知 $F_0 \leq 30\text{Hz}$ $f_{\text{H}} = 1\text{kHz}$

(1) $T_{\text{pmin}} = \frac{1}{F_{\text{max}}} = \frac{1}{50} = 0.02\text{s}$

(2) $T_{\text{max}} = \frac{1}{f_{\text{S}}} = \frac{1}{2f_{\text{H}}} = \frac{1}{2\text{kHz}} = 0.5\text{ms}$

(3) $N_{\text{min}} = \frac{T_{\text{pmin}}}{T_{\text{max}}} = 0.04 \times 10^3 = 40$

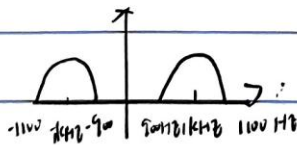
(4) $F_0 = \frac{1}{N T_{\text{max}}}$

f_{H} 不变 T_{max} 不变

为使 F_0 减小一半

$$N^2 = 2N_{\text{min}} = 80$$

3-19 $f_c = 1\text{kHz}$ $f_m = 100\text{Hz}$



为了保证传输信息足够多

则 $F \leq 100\text{Hz}$

且 N 为整数, 否则会丢失部分信息.



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$$(1) \quad T_{pmin} = \frac{1}{F} = \frac{1}{100} = 0.01s$$

$$(2) \quad f_{smin} = 2f_{max} = 2 \times 1.1kHz = 2.2kHz$$

基于奈奎斯特采样定理

$$(3) \quad N_{min} = \frac{T_{pmin}}{T_{max}} = T_{pmin} \times 2f_{max} = 22$$

因为为FFT 采样点数应为2^M次方

则最少采样点数为32